RELATED RATES

DIFFERENTIAL CALCULUS

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

60%

EXAM

40%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

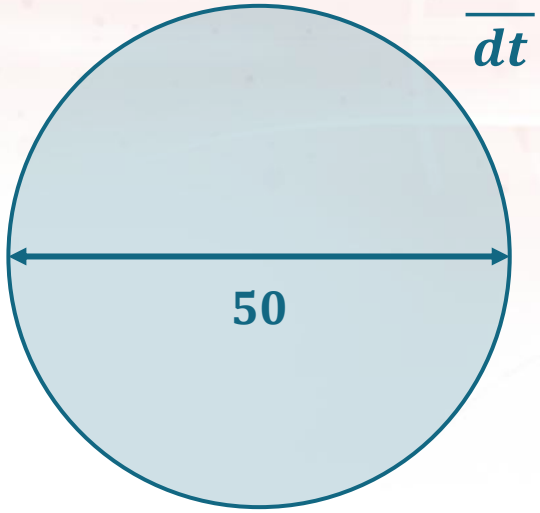
Related Rates Problem Solving

RELATED RATES

PROBLEM SOLVING

Air is being pumped into a spherical ballon so that its **volume** increases at a rate of $\mathbf{100\ cm^3/s}$. How **fast** is the **radius** of the ballon increasing when the **diameter** is $\mathbf{50\ cm}$?

$$\frac{dV}{dt} = 100$$



$$\left.\frac{dr}{dt}\right|_{r=25}$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dt} = \frac{4}{3}\pi \frac{d}{dt} r^3$$

$$\frac{dV}{dt} = \frac{4}{3}\pi \frac{d}{dr} r^3 \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = \frac{4}{3}\pi (3r^2) \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = 4\pi r^2 \cdot \frac{dr}{dt}$$

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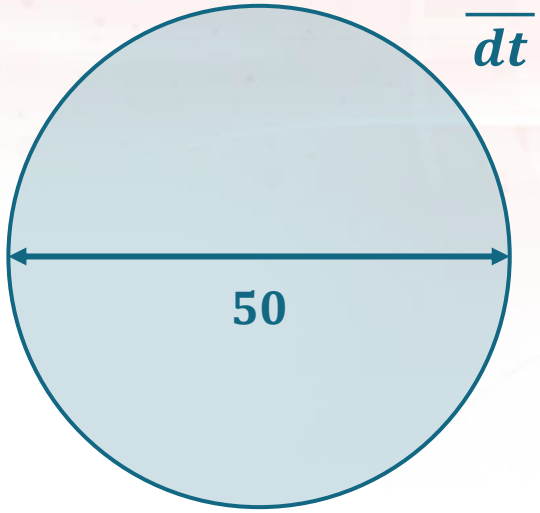


$$\frac{dV}{dt} = 100$$

$$\left.\frac{dr}{dt}\right|_{r=25}$$

$$\frac{dr}{dt} = \frac{dV}{dt} \cdot \frac{1}{4\pi r^2}$$

$$\frac{dr}{dt} = 100 \cdot \frac{1}{4\pi r^2}$$

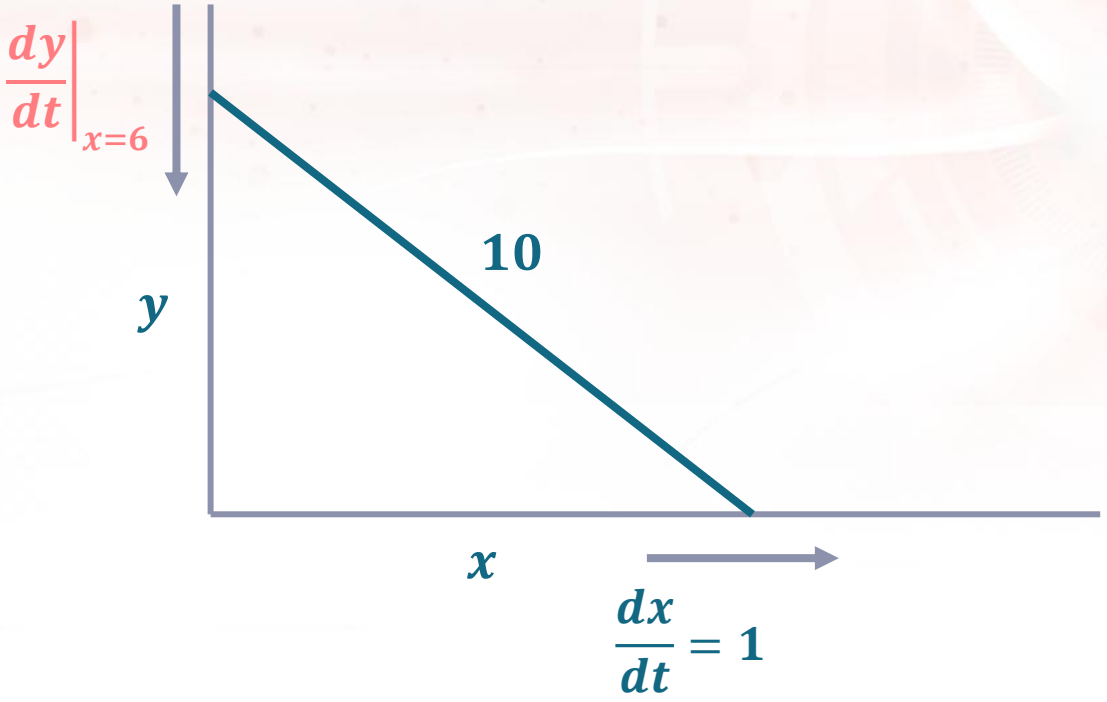
$$\frac{dr}{dt} = \frac{25}{\pi r^2}$$

$$\left.\frac{dr}{dt}\right|_{r=25} = \frac{25}{\pi(25)^2}$$

$$\boxed{\left.\frac{dr}{dt}\right|_{r=25} = \frac{1}{25\pi}\ cm/s}$$

ans

A ladder **10 ft** long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of **1 ft/s**, how **fast** is the **top** of the ladder sliding down the wall when the bottom of the ladder is **6 ft** from the wall?



$$x^2 + y^2 = 100$$

$$\frac{d}{dt}x^2 + \frac{d}{dt}y^2 = \frac{d}{dt}100$$

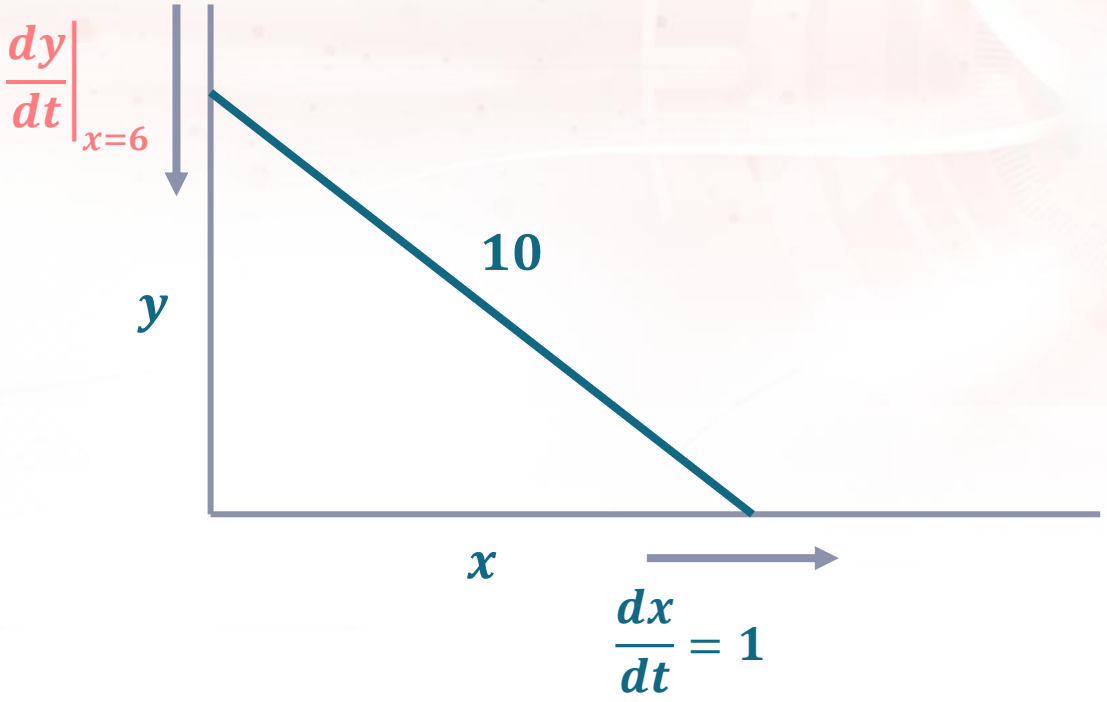
$$\frac{d}{dx}x^2\frac{dx}{dt} + \frac{d}{dy}y^2\frac{dy}{dt} = 0$$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$2y\frac{dy}{dt} = -2x\frac{dx}{dt}$$

$$\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$

A ladder **10 ft** long rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of **1 ft/s**, how **fast** is the **top** of the ladder sliding down the wall when the bottom of the ladder is **6 ft** from the wall?



Let $x = 6$

$$6^2 + y^2 = 100$$

$$y = \sqrt{100 - 36}$$

$$y = 8$$

$$\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$

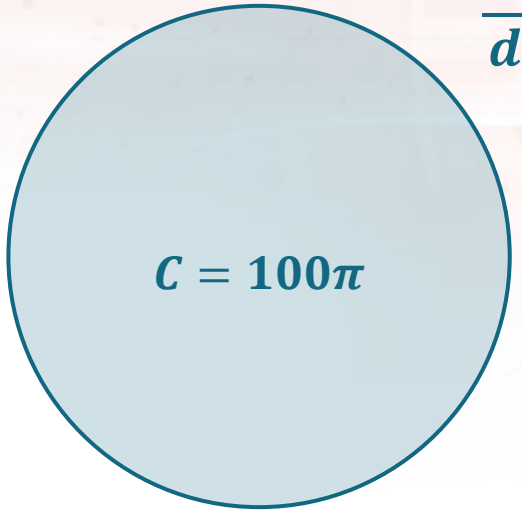
$$\left.\frac{dx}{dt}\right|_{x=6} = -\frac{6}{8}(1)$$

$$\boxed{\left.\frac{dx}{dt}\right|_{x=6} = -\frac{3}{4}\ ft/s}$$

ans

Oil spilled from a tanker spreads in a circle whose circumference increases at a rate of $\mathbf{40\ ft/s}$. How **fast** is the **area** of the spill increasing when the circumference of the circle is $\mathbf{100\pi\ ft}$?

$$\frac{dC}{dt} = 40$$



$$\left.\frac{dA}{dt}\right|_{C=100\pi}$$

$$C = 2\pi r \qquad A = \pi r^2$$

$$r = \frac{C}{2\pi}$$

$$A = \pi\left(\frac{C}{2\pi}\right)^2$$

$$A = \pi\frac{C^2}{4\pi^2}$$

$$A = \frac{C^2}{4\pi}$$

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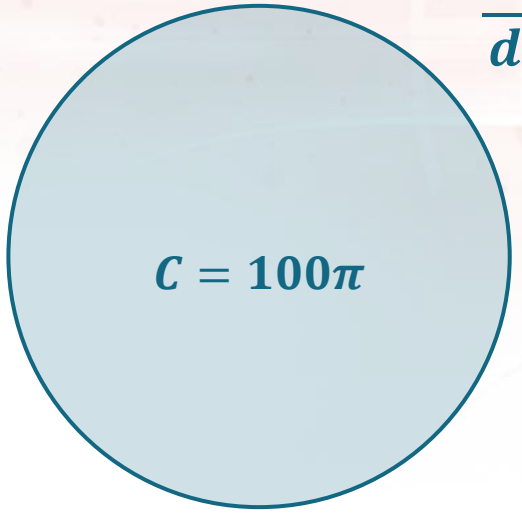


$$\left.\frac{dA}{dt}\right|_{C=100\pi}$$

$$\frac{dA}{dt} = \frac{d}{dt}\left(\frac{C^2}{4\pi}\right)$$

$$\frac{dA}{dt} = \frac{1}{4\pi}\frac{d}{dC}(C^2)\frac{dC}{dt}$$

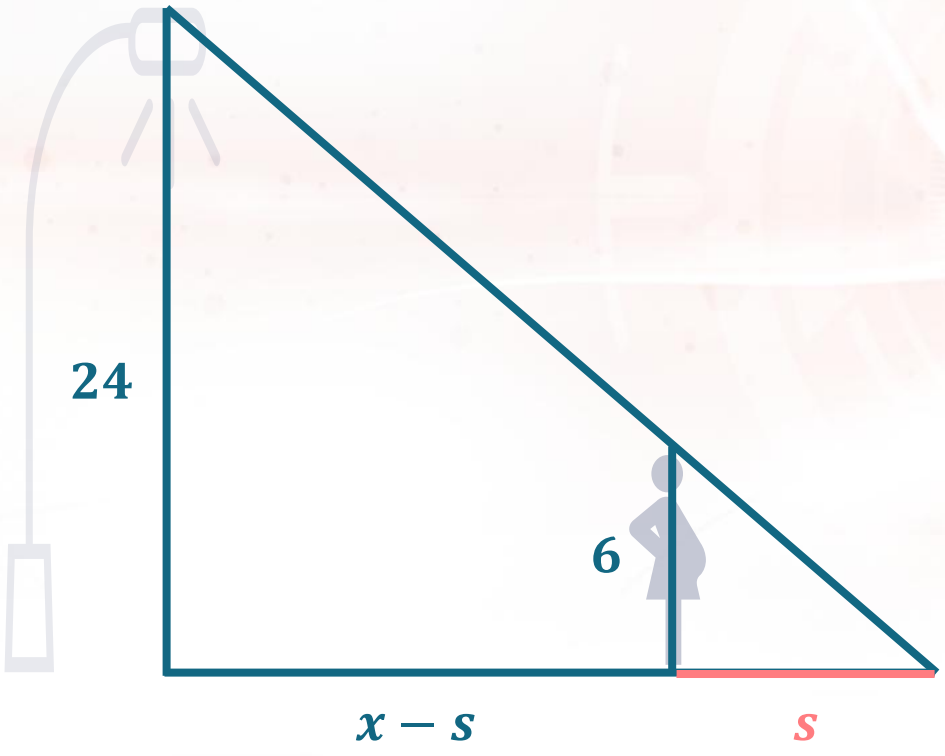
$$\frac{dA}{dt} = \frac{1}{4\pi}2C\frac{dC}{dt}$$

$$\frac{dA}{dt} = \frac{1}{4\pi}2(100\pi)(40)$$

$$\boxed{\frac{dA}{dt} = 2000\ sq.ft./s}$$

ans

A **6-ft** tall woman is walking at the rate of $\mathbf{4\ ft/s}$ away from a streetlamp that is $\mathbf{24\ ft}$ tall. How **fast** is the length of her **shadow** changing?



Similar Triangles

$$\frac{24}{x-s} = \frac{6}{s}$$

$$24s = 6x - 6s$$

$$18s = 6x$$

$$s = \frac{1}{3}x$$

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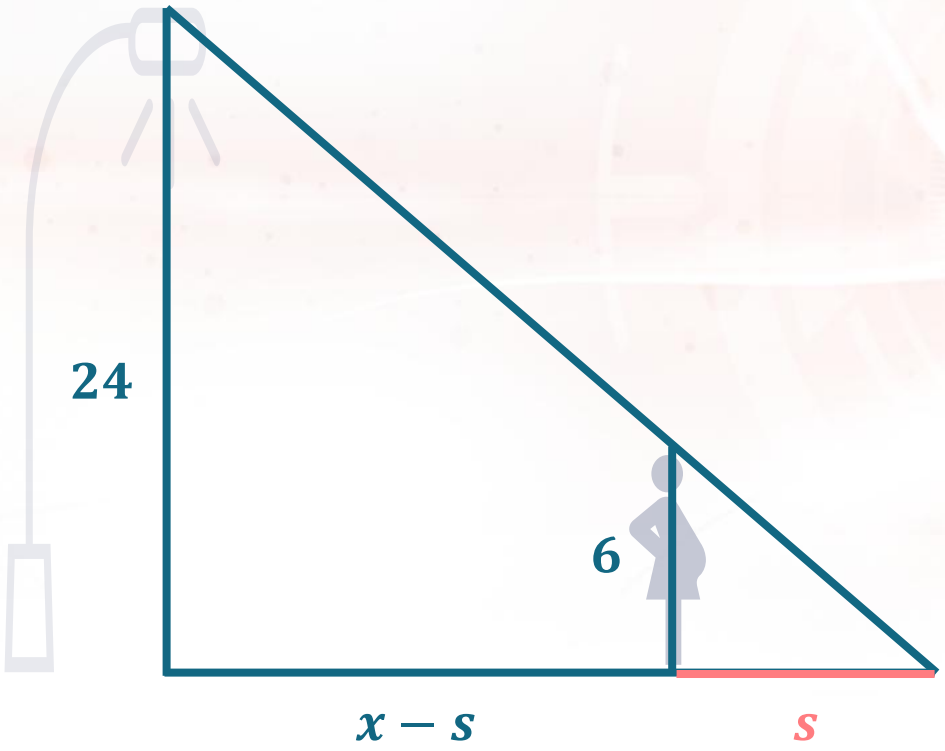

$$\frac{ds}{dt} = \frac{1}{3}\frac{d}{dt}x$$

$$\frac{ds}{dt} = \frac{1}{3}\frac{d}{dx}x \cdot \frac{dx}{dt}$$

$$\frac{ds}{dt} = \frac{1}{3}\frac{dx}{dt}$$

$$\frac{ds}{dt} = \frac{1}{3}(4)$$

$$\boxed{\frac{ds}{dt} = \frac{4}{3}\ ft/s}$$

ans

SEATWORK